

3D-MOT candidate selection and optimization

Executive summary

Decision. Fully optimize the six-blue-beam donut as the demonstrated performance reference and the two-blue single pass as the simpler laboratory alternative. Pages 1-2 summarize the decision and deliverables; pages 3-7 provide supporting evidence and methodology.

Candidate	Why retain it	Main limitation	Status
Full donut	Best demonstrated capture; 491/600 usable in paired screening	Six blue beams and greater optical complexity	Ready for 7D optimization
Two-blue single pass	Only two localized blue beams; earlier tuning reached 115/600 in a related geometry	New yz geometry is not yet performance-validated	Run short screen, then 9D optimization if viable

Final result for each geometry

Recommended settings; mean conditional efficiency X with a 95% confidence interval; expected 3D-MOT atomic flux; and an empirically supported operating box in the **7D donut or 9D single-pass optimized parameter space**. The parameter definitions are listed on page 2.

Primary metric

An atom is usable at 100 ms when its distance from the MOT center is at most 5 mm and its total speed is at most 1 m/s. The reported percentage is conditional on the atoms that survived the 2D MOT. The predicted 2D-MOT output is $6.63701\text{e}9$ - $6.92764\text{e}9$ atoms/s, so the final conditional result will also be converted to an expected 3D-MOT atomic flux.

What will be optimized and reported

Family	Optimized parameters	Dimension
Full donut	green s0, detuning, waist; blue s0, detuning, waist; magnetic gradient	7D
Two-blue single pass	same seven parameters; included blue-beam angle; crossing z position	9D

The six green MOT beams retain the fixed 60/120-degree geometry. In the single pass, two blue beams lie in the yz plane, cross upstream of the MOT, and propagate with negative-z components. The seed geometry is 45 degrees at $z = -10$ mm; angle is restricted to 45-70 degrees. Blue s0 is capped at 1.5.

High-level workflow

1. Validate conventions and time a short run. 2. Screen the new yz single pass on the same 600 atoms used for paired comparisons. 3. Run resumable Optuna discovery. 4. Refine finalists on larger, independent existing ensembles. 5. Construct and validate a near-optimal operating box. 6. Freeze all choices and evaluate the final claim on newly generated sealed ensembles.

The existing 15,840 survivors are used for final candidate selection, not for the final unbiased performance claim. Final reported efficiency is evaluated only on newly generated sealed ensembles.

Questions to confirm before production

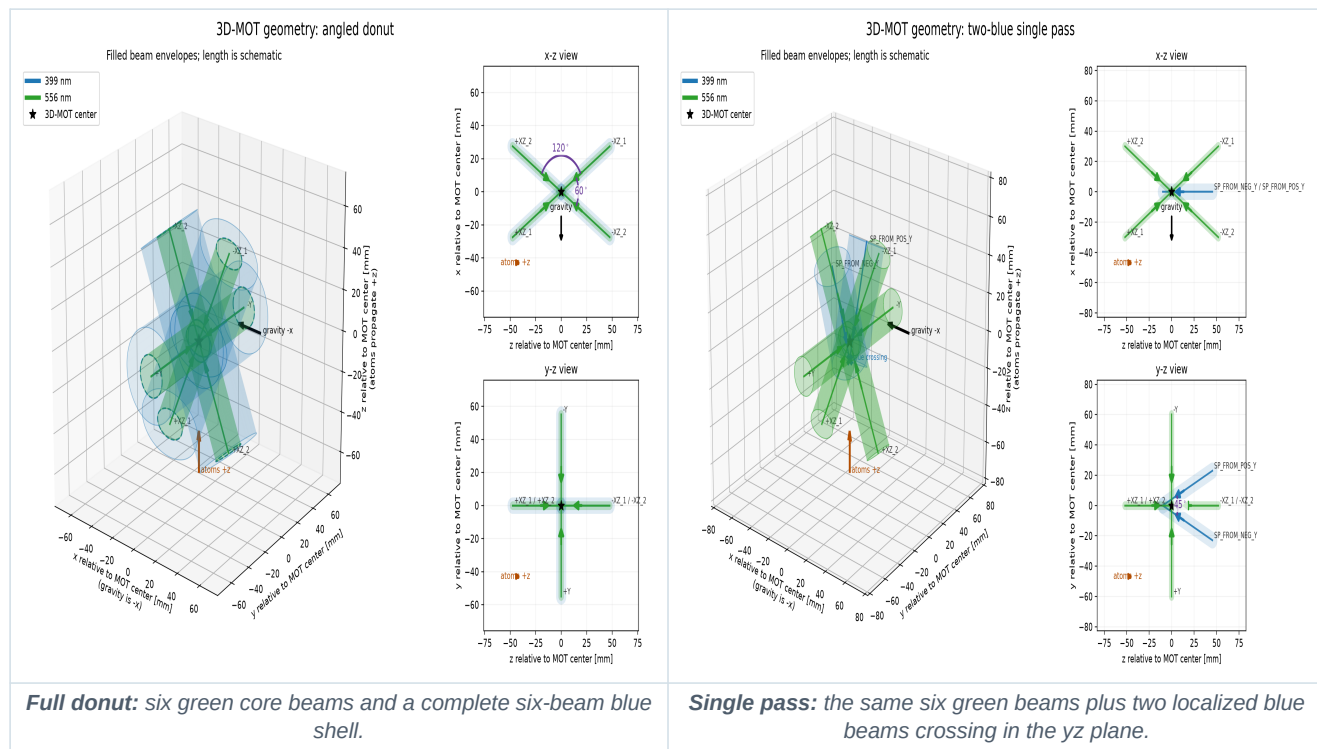
- Laboratory bounds, resolution, drift, and available power for each optimized control.
- Mechanically accessible z range and resolution for the single-pass crossing.
- Accepted negligible-loss threshold for the operating box.
- Whether controls are shared within each wavelength group, as assumed.

Planning budget

Stage	Maximum evaluations	Particles x seeds	Node-h
Implementation	24 diagnostics	2-600 x 1	6
Discovery	350 donut + 600 single pass	600 x 1	216
Refinement + finalists	16 + 3 / family	3,000 x 3; 15,840 x 3	504
Robust box	69 + 8 / family	1,800 x 1; 3,000 x 3	360
Checks + final report	12 / family + 2 nominal	600 x 3; about 16,000 x 3	216

Pre-extension planning ceiling: 1,302 node-hours, including 6 node-hours for implementation. One optional extension of at most 100 additional donut trials or 150 additional single-pass trials is permitted only after a stop-or-approve checkpoint and must fit a separately approved 72-node-hour cap.

The two candidate geometries



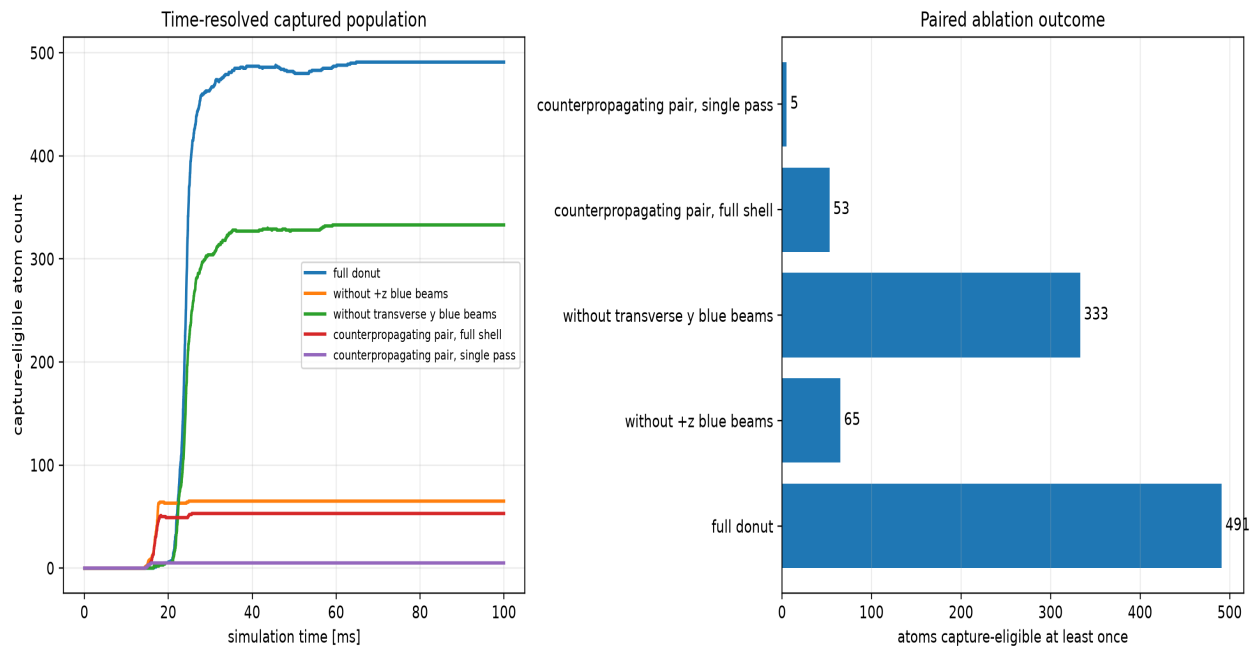
The new single-pass geometry is deliberately simple: two blue beams provide the entrance slowing before the atoms reach the green MOT. Both have negative- z propagation components; their included angle and common crossing position are optimization variables.

Short-screen gate

Because earlier single-pass tuning used a different beam plane, its 115/600 result is motivation rather than validation. The new yz implementation first receives a paired 600-atom screen. Roughly 20% usable capture is sufficient to justify the full search; a much lower result triggers a geometry and sign-convention audit before expensive allocation.

Why the complete donut remains the reference

Which 399-nm beam functions make the donut effective?



Paired ablations use the same 600 incoming atoms, so the differences isolate the role of blue-beam groups.

Configuration	Usable	Fraction
Full donut	491/600	81.8%
Without +z blue pair	65/600	10.8%
Without transverse-y pair	333/600	55.5%
One pair, continuous shell	53/600	8.8%
One pair, initial single pass	5/600	0.8%

Interpretation. Transverse cooling, forces in both longitudinal senses, and repeated blue-light access all matter. Removing blue directions strongly reduces capture, so the full donut is the appropriate performance reference.

Why a two-blue single pass is still worth optimizing

Earlier localized-pair evidence: 5/600 at the initial point; 52/600 after tuning blue s0 and detuning; 115/600 (19.2%) after jointly tuning blue s0, detuning, and gate position. A four-blue entrance-plus-return endpoint reached 188/600 (31.3%), still far below the donut.

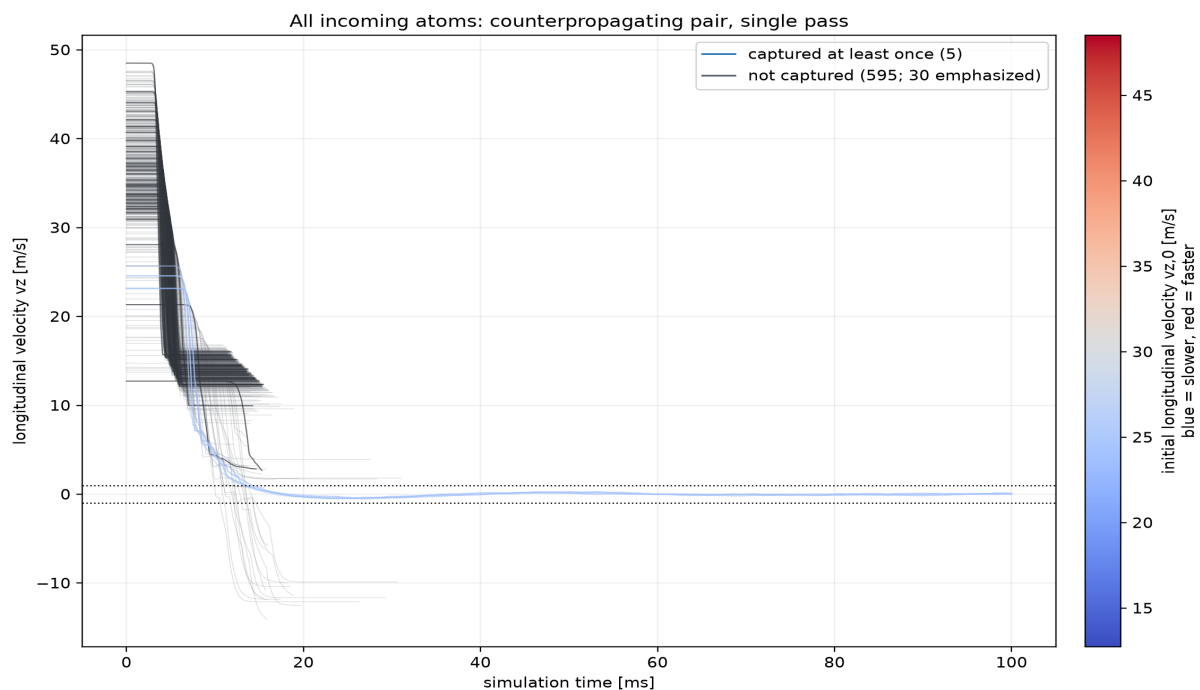
This history shows two things. First, a localized blue pair is highly sensitive to its operating point and can improve substantially under tuning. Second, adding a return pair does not recover donut-level performance. The main optimization budget is therefore better spent comparing the full donut with the genuinely simpler two-blue architecture, rather than optimizing an intermediate four-blue system.

What is new

The production candidate moves the blue pair to the yz plane and makes the included angle an explicit variable. Its two beams originate on opposite y sides and propagate toward a shared upstream crossing. This is the geometry the laboratory judged easier to construct.

What is not yet claimed

We do not assign the earlier 19.2% result to the new geometry. Until the paired screen is complete, the new candidate has geometry validation but no demonstrated capture figure. This prevents historical evidence from being presented as an unbiased result for a different configuration.



Earlier single-pair trajectory evidence illustrates the physical slowing mechanism; it is not the final yz-candidate performance result.

Uncertainty, selection, and unbiased validation

Discovery trials use common initial atoms and matched recoil seeds, making nearby candidates comparable with reduced Monte Carlo noise. Existing 2D-MOT survivor ensembles are separated between discovery, refinement, and an early independent check.

Data	Role	May determine parameters?	Supports final claim?
Existing discovery ensembles	Optuna and screening	Yes	No
Existing refinement/check ensembles	Finalist and box selection	Yes	No
New sealed ensembles	Final locked-point evaluation	No	Yes

Confidence interval

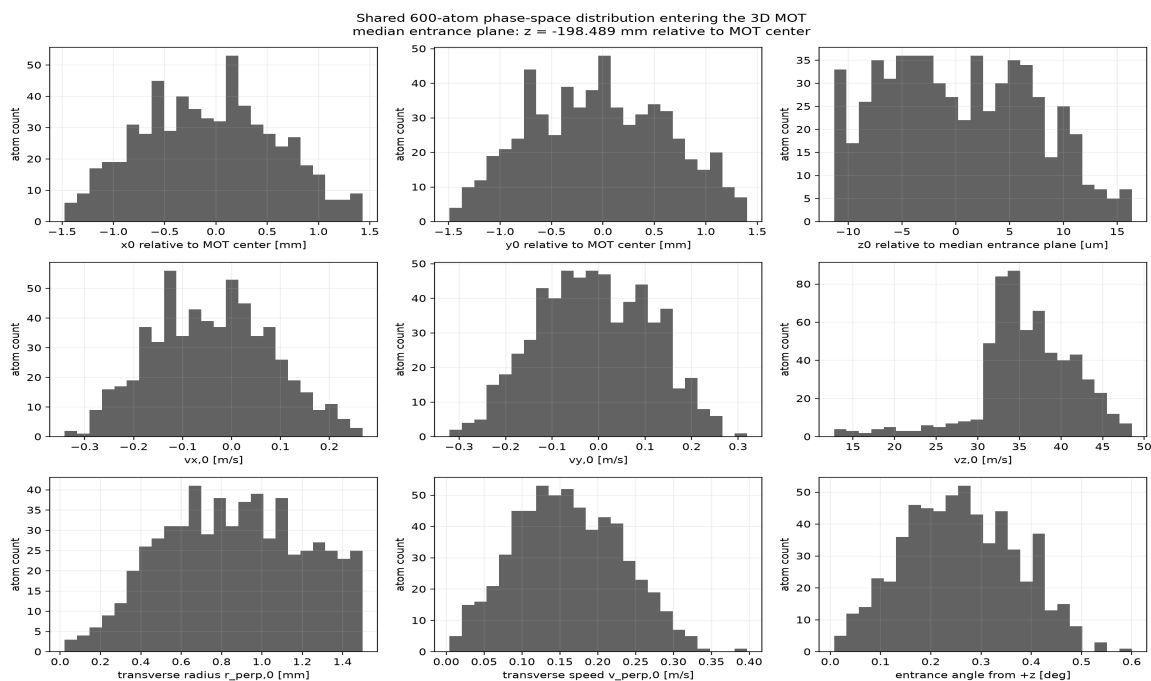
The target is a 95% confidence-interval half-width no larger than 0.75 percentage points for the mean conditional capture efficiency. Recoil-seed realizations are crossed across ensemble groups. A crossed hierarchical bootstrap independently resamples 2D-MOT ensemble clusters and recoil-seed realizations while preserving particle grouping within each ensemble.

Close finalists

Paired finalist differences target a 95% half-width no larger than 0.4 percentage points. If two settings remain statistically tied, prefer the lower-power, more interior, and less sensitive point rather than claiming a numerically unique optimum.

Fraction versus percent

Calculations use fractions internally. Reported efficiencies are converted to percentages and uncertainty is stated in percentage points. Atomic flux is reported separately in atoms/s, not as an atom count.



The shared entrance ensemble used for paired geometry comparisons.

From a best point to a usable laboratory recommendation

A single numerical optimum is not enough. Around each locked nominal point, the campaign constructs the largest empirically supported axis-aligned box whose sampled worst points remain within the laboratory-approved negligible-loss threshold.

Deliverable	Meaning
Nominal settings	Recommended center point with every optimized control listed
Efficiency $X \pm Y$	Mean conditional capture with a 95% confidence interval
Expected atomic flux	Conditional result combined with the independent 2D-MOT output
7D or 9D operating box	Per-parameter ranges supported jointly, not only one at a time
Diagnostics	Population history, entrance phase space, trajectories, and retention checks

Operating-box construction

Start with one-dimensional sensitivity scans, test joint perturbations and corners, shrink dimensions that fail, and validate the apparent worst points with larger particle samples and multiple recoil seeds. A box is reported only if its joint samples satisfy the declared loss threshold.

Stop rules

The campaign is resumable and budget-capped. Discovery receives at most one approved extension after a formal checkpoint. Statistical validation is not silently reduced to create more Optuna trials. If the single-pass short screen fails, the full 9D allocation pauses rather than optimizing a likely geometry error.

Outcome for the laboratory. The comparison will quantify the performance and robustness gained by the full donut against the optical simplicity of the two-blue single pass. Both recommendations will be expressed in experimentally adjustable parameters with uncertainty and tolerance ranges, enabling a transparent engineering choice.

Detailed sources

The complete variable domains, data partitions, budget limits, stopping rules, and validation procedure are maintained in docs/3D_MOT_FULL_OPTIMIZATION_PLAN.md. Candidate evidence and figure links are maintained in docs/3D_MOT_OPTIMIZATION_CANDIDATE_RATIONALE.md.